Mehrab Uddin Ahmed

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PROFILE

AI/ML enthusiast with around one year of hands-on experience working on machine learning models and IoT-based solutions, especially in the agriculture sector. I have worked with data preprocessing, feature engineering, and basic deep learning techniques to solve real-world problems.

I enjoy working in a team, learning new technologies, and applying my skills to build practical and impactful solutions. I am always eager to improve myself and use data to create smart, efficient systems that can make a real difference.

EMPLOYMENT HISTORY

Jan 2024 — Present

Research Assistant, North South University — Dhaka

- Developing and implementing machine learning models for IoT-enabled crop recommendation systems, enhancing agricultural productivity and sustainability.
- Collaborating with interdisciplinary teams to integrate IoT devices and cloud platforms for data collection, storage, and analysis.
- Preprocessing large datasets, including image data, to train and validate convolutional neural network (CNN) models for disease detection.

Jun 2024 — Dec 2024

Research Assistant, Habiganj Agricultural University — Dhaka

- Conducted research on precision agriculture, leveraging IoT-based solutions to optimize crop management and resource allocation.
- Developed and implemented machine learning models for IoT-enabled crop recommendation systems, enhancing agricultural productivity and sustainability.
- Designed and deployed an IoT system to collect real-time environmental data for monitoring soil, weather, and crop conditions.
- Collaborated with interdisciplinary teams to integrate IoT devices and cloud platforms for data collection, storage, and analysis.
- Led a project on Malabar Spinach disease detection, utilizing Deep learning techniques to classify and predict plant diseases accurately.
- Preprocessed large datasets, including image data, to train and validate convolutional neural network (CNN) models for disease detection.
- Conducted field experiments and collected data to evaluate the effectiveness of IoT and ML solutions in improving crop yield and health

Mar 2025 — Aug 2025

AI/ML Engineer Intern, Gaze Technology — Dhaka

- Assisted with data cleaning, feature engineering, and testing of CV models for security analytics.
- Participated in Agile sprints; used Git/GitHub for version control.
- Contributed to prototype deployment and performance monitoring, improving model reliability.

EDUCATION

Jan 2020 — Dec 2024

B.Sc in Computer Science and Engineering, North South University — Dhaka

Apr 2017 — Apr 2019

Science, Brahmanbaria Govt. College — Brahmanbaria

Jan 2012 — Apr 2017

Science, Annada Govt. High School — Brahmanbaria

SKILLS

Machine Learning(Supervised & Unsupervised Learning- Classification, Regression, Clustering Model training & evaluation (accuracy, precision, recall, F1-score))

Deep Learning(Neural Networks (basics) CNN fundamentals-

AI Tools & Libraries(Jupyter Notebook OpenCV (basic image processing) Git/GitHub (version control))

Problem Solving & API Integration

MS Office

Frameworks: TensorFlow / Keras (beginner), PyTorch (beginner))

Data Science (Data preprocessing (cleaning, normalization, encoding)- Feature engineering Exploratory Data Analysis (EDA))

Jul 2025

Publications

Dhaka

An IoT-enabled AI system for real-time crop prediction using soil and weather data in precision agriculture

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LANGUAGES

English — Good working knowledge

Bangla — Native speaker

REFERENCES

Dr. Riasat Khan from North South University

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